

Rounding, Estimation and Bounds



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A rounded number hides a range: half the unit below, half the unit above. For the biggest answer, choose each bound to push the answer up.

THE TOPIC ON ONE MAP

Rounding, Estimation and Bounds

01 Rounding and error intervals

4.3 kg to 2 s.f.

$\dots \leq L < \dots$

18 000 to the nearest 1000

truncated: $8.3 \leq y < 8.4$

02 Estimation

each number to 1 s.f.

is his answer sensible?

over- or underestimate?

03 Bounds in a formula

upper bound of $\frac{2a-c}{d}$

lower bound of $\frac{t}{a-h}$

sine rule with bounds

04 Bounds in context: decide

could the speed be over 80?

enough soup?

"to a suitable degree of accuracy"

HALF THE UNIT The rounded value sits in the middle of its interval:



01 Rounding and Error Intervals

Find the unit of the rounding, halve it, then go that far down and up. The lower bound is included ($\leq$), the upper bound is not ($<$).

WHEN YOU SEE → YOU DO

“correct to 1 d.p.” / “to 2 s.f.” / “to the nearest 5”

→ unit 0.1 / last significant place / 5 → halve it

“write down the error interval”

→ $\text{lower} \leq x < \text{upper}$

“minimum / maximum number of people”

→ whole numbers: 17 500 and 18 499

“truncated” / “the display began 8.3”

→ $8.3 \leq y < 8.4$ (not half the unit)

a perimeter from rounded sides

→ multiply both bounds

Find the unit

Halve it

– half : lower

+ half : upper

WORKED MODEL • JUNE 2025 P1HR

A ship is 142.8 m, correct to 1 decimal place. Write down the bounds.

1 unit 0.1, half 0.05

1 d.p.

ANSWER $142.75 \leq L < 142.85$

× **TRAP** 142.84 is not the upper bound. The upper bound is 142.85 — the value it cannot reach.

IDEA 01

Practise: Rounding and Error Intervals

My questions first, then selected practice, then real past papers.

Q1

DR ESLAM

Half the unit

CH 1 • TOPIC 9

2 marks

$w = 6.4$ kg correct to 1 decimal place. Write the error interval for w .

SELECTED PRACTICE

Q2

PRACTICE

Practice • Easy

CH 1 • TOPIC 9

2 marks

- 1 A number, n , is rounded to 2 decimal places.
The result is 4.76
Using inequalities, write down the error interval for n .

(2 marks)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 9**2 marks**

7 The length of a book is 33.8 cm, correct to one decimal place.

i) Write down the lower bound of the length of the book.

..... cm [1]

ii) Write down the upper bound of the length of the book.

..... cm [1]

(2 marks)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 9**3 marks**

12 (a) The length of each side of a regular pentagon is 8.4 cm to 1 decimal place.

Complete the error interval for the length of one side.

.....cm $\leq$ length <cm

(2 marks)

(b) Complete the error interval for the perimeter.

.....cm $\leq$ perimeter <cm

(1 mark)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 9 4 marks

13 (a) The length of a pipe is 6 metres to the nearest metre.

Complete the error interval for the length of the pipe.

.....m $\leq$ length <m

(2 marks)

(b) The length of a different pipe is 4 metres to the nearest metre.

Olly says,

“The total length of the two pipes is 11 metres to the nearest metre.”

Give an example to show that he could be correct.

(2 marks)

PAST PAPERS

Q6**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q11

CH 1 • TOPIC 9 4 marks

11 The weight of a cat is 4.3 kg correct to 2 significant figures.

(a) Write down the upper bound of the weight of the cat.

..... kg

(b) Write down the lower bound of the weight of the cat.

..... kg

$$G = e - f$$

$e = 17$ correct to the nearest integer

$f = 9.4$ correct to one decimal place

(c) Work out the upper bound for the value of G .

.....

(Total for Question 11 is 4 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q4

CH 1 • TOPIC 9 2 marks

- 4 Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

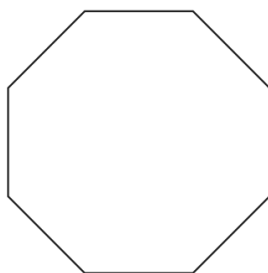


Diagram **NOT**
accurately drawn

- (a) Write down the lower bound of the length of each side of the octagon.

..... mm

- (b) Write down the upper bound of the length of each side of the octagon.

..... mm

(Total for Question 4 is 2 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q2

CH 1 • TOPIC 9 2 marks

- 2 The length of a table is measured as 1.4 metres correct to one decimal place.

- (a) Write down the upper bound of the length of the table.

..... metres

- (b) Write down the lower bound of the length of the table.

..... metres

(Total for Question 2 is 2 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q2

CH 1 • TOPIC 9 2 marks

2 The length of a ship is 142.8 m, correct to 1 decimal place.

(i) Write down the lower bound of the length of the ship.

..... m

(ii) Write down the upper bound of the length of the ship.

..... m

(Total for Question 2 is 2 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q7

CH 1 • TOPIC 9 2 marks

7 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm

(Total for Question 7 is 2 marks)

02 Estimation

Round every number to 1 significant figure, then calculate without a calculator. For a square root, round to a nearby square number.

WHEN YOU SEE → YOU DO

“work out an estimate”

→ each number to 1 s.f., show the rounded line

“check his answer is sensible”

→ estimate, then compare the size

$\sqrt{63.5 \times 101.7}$

→ $\sqrt{64 \times 100} = 80$ (nice squares)

“overestimate or underestimate?”

→ look at which numbers you rounded up / down

WORKED MODEL • NOVEMBER 2025 P1H

Estimate $\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$.

1 $\frac{2^2 \times 60}{\sqrt{9}}$

1 s.f. (and a square)

2 $\frac{240}{3}$

calculate

ANSWER 80

OVER OR UNDER? Top rounded up or bottom rounded down → the estimate is too big (overestimate).

IDEA 02

Practise: Estimation

My questions first, then selected practice, then real past papers.

Q11 DR ESLAM 1 s.f.

CH 1 • TOPIC 9 2 marks

Estimate $\frac{49.6 \times 3.82}{0.198}$.

SELECTED PRACTICE**Q12** PRACTICE Practice • Easy

CH 1 • TOPIC 9 3 marks

14 Asha worked out $\frac{326.8 \times (6.94 - 3.4)}{59.4}$

She got an answer of 19.5, correct to 3 significant figures.

Write each number correct to 1 significant figure to decide if Asha's answer is reasonable.

(3 marks)

Q13**PRACTICE**

Practice • Medium

CH 1 • TOPIC 9 2 marks

- 10 Nav has worked out $\frac{68.3 \times 42.8}{0.021}$ on his calculator.

His answer is 139 201.9048

Without using a calculator and using suitable approximations, check that his answer is sensible.

Show your working clearly.

(2 marks)

Q14**PRACTICE**

Practice • Hard

CH 1 • TOPIC 9 4 marks

10

Competition
a prize every 2014 seconds

In a competition, a prize is won every 2014 seconds.

Work out an estimate for the number of prizes won in 24 hours. You must show your working.

(4 marks)

Q15**PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 9 4 marks

- 6 (a) One uranium atom has a mass of 3.95×10^{-22} grams.

Work out an estimate for the number of uranium atoms in 1 kg of uranium.

(3 marks)

- (b) Is your answer to (a) an underestimate or an overestimate?

Give a reason for your answer.

(1 mark)**PAST PAPERS****Q16****PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q2

CH 1 · TOPIC 9 2 marks

- 2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

(Total for Question 2 is 2 marks)

03 Bounds in a Formula

To make an answer as **big** as possible: add the big ones, subtract the small one, multiply by the big one, divide by the small one.

WHEN YOU SEE → YOU DO

“upper bound of $a + b / a \times b$ ”

→ upper + upper, upper × upper

“upper bound of $a - b$ ”

→ upper - **lower**

“upper bound of $\frac{a}{b}$ ”

→ upper ÷ **lower**

a formula such as $\frac{2a - c}{d}$

→ decide each letter one at a time

“show your working clearly”

→ write the bound you chose for every letter

ZONE BIGGEST

Upper bound of the answer

$$\frac{\text{UP} + \text{UP}}{\text{LOW}}, \quad \text{UP} - \text{LOW}$$

push up everything on top, down everything underneath

ZONE SMALLEST

Lower bound of the answer

$$\frac{\text{LOW} + \text{LOW}}{\text{UP}}, \quad \text{LOW} - \text{UP}$$

the opposite of every choice

WORKED MODEL • JANUARY 2020 P1H

$P = \frac{2a - c}{d}$, $a = 58.4$ (3 s.f.), $c = 20$ (2 s.f.), $d = 3.6$ (2 s.f.). Upper bound of P ?

1 $a = 58.45, c = 19.5, d = 3.55$

up · low · low

2
$$\frac{2(58.45) - 19.5}{3.55}$$

substitute

ANSWER 27.44

× **TRAP** Using upper - upper for the upper bound of a subtraction. Subtract the **smallest** possible amount.

IDEA 03

Practise: Bounds in a Formula

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Biggest

CH 1 • TOPIC 9 3 marks

$x = 12$ (nearest whole), $y = 3.4$ (1 d.p.). Find the upper bound of $\frac{x}{y}$.

SELECTED PRACTICE

Q18**PRACTICE**

Practice • Medium

CH 1 • TOPIC 9 4 marks

- 1 (a) The value of p is 4.3
The value of q is 0.4

Both p and q are given correct to the nearest 0.1

Write down the lower bound for p .

(1 mark)

(b) $r = p + \frac{1}{q}$

Work out the upper bound for r .
You must show all your working.

(3 marks)

Q19**PRACTICE**

Practice · Medium

CH 1 · TOPIC 9 3 marks

$$4 \quad D = \frac{x}{y}$$

 $x = 99.7$ correct to 1 decimal place. $y = 67$ correct to 2 significant figures.Work out an upper bound for D .

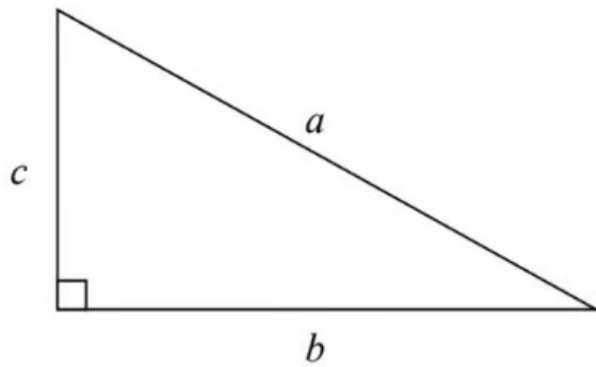
(3 marks)

Q20**PRACTICE**

Practice · Hard

CH 1 · TOPIC 9 4 marks

4



a is 8.3 cm correct to the nearest mm

b is 6.1 cm correct to the nearest mm

Calculate the upper bound for c .

You must show your working.

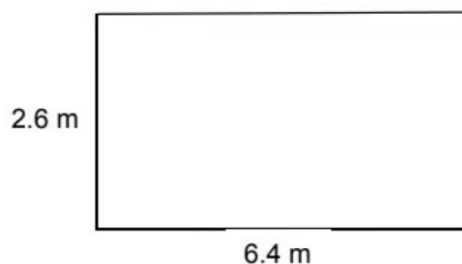
(4 marks)

Q21**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 9 5 marks

- 7 The dimensions of a rectangular floor are to the nearest 0.1 metres.



Not drawn
accurately

A force of 345 Newtons is applied to the floor.

The force is to the nearest 5 Newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the upper bound of the pressure.

Give your answer to 4 significant figures.

You **must** show your working.

.....N/m² [5]

(5 marks)

PAST PAPERS

Q22 **PAST PAPER** Edexcel 4MA1 • January 2020 • Paper 1H • Q16**CH 1 • TOPIC 9** **3 marks**

16 $P = \frac{2a - c}{d}$

$a = 58.4$ correct to 3 significant figures.

$c = 20$ correct to 2 significant figures.

$d = 3.6$ correct to 2 significant figures.

Work out the upper bound for the value of P .

Show your working clearly.

Give your answer correct to 2 decimal places.

.....

(Total for Question 16 is 3 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q19

CH 1 • TOPIC 9 3 marks

19 The diagram shows rectangle $ABCD$ with rectangle $EFGH$ cut out to form the shaded region.

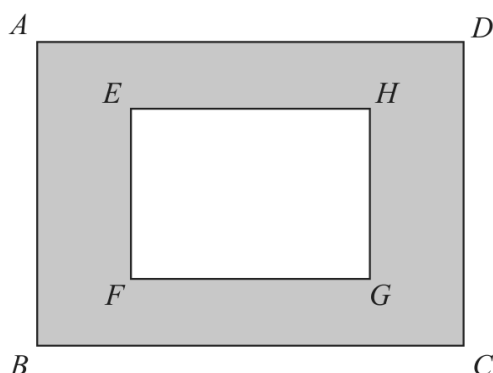


Diagram **NOT**
accurately drawn

$AD = 8.3$ cm correct to one decimal place

$DC = 7.2$ cm correct to one decimal place

$EH = 6.2$ cm correct to one decimal place

$HG = 5.3$ cm correct to one decimal place

Work out the upper bound of the area of the shaded region.

Show your working clearly.

..... cm^2

(Total for Question 19 is 3 marks)

Q24 **PAST PAPER** Edexcel 4MA1 • January 2021 • Paper 1H • Q15**CH 1 • TOPIC 9** **3 marks****15**

$$a = \frac{v - u}{t}$$

 $v = 9.6$ correct to 1 decimal place $u = 3.8$ correct to 1 decimal place $t = 1.84$ correct to 2 decimal placesCalculate the upper bound for the value of a .

Give your answer as a decimal correct to 2 decimal places.

Show your working clearly.

.....

(Total for Question 15 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q19

CH 1 • TOPIC 9 3 marks

19 $k = \frac{t}{a - h}$

 $t = 14$ correct to 2 significant figures $a = 7.8$ correct to 2 significant figures $h = 3.4$ correct to 2 significant figuresWork out the lower bound for the value of k .

Show your working clearly.

.....

(Total for Question 19 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q20

CH 1 • TOPIC 9 3 marks

20 $P = \frac{t - w}{y}$

$t = 9.7$ correct to 1 decimal place

$w = 5.9$ correct to 1 decimal place

$y = 3$ correct to 1 significant figure

Calculate the upper bound for the value of P .
Show your working clearly.

.....
(Total for Question 20 is 3 marks)

Q27**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q13

CH 1 • TOPIC 9 3 marks

13 Here is a triangle XYZ .

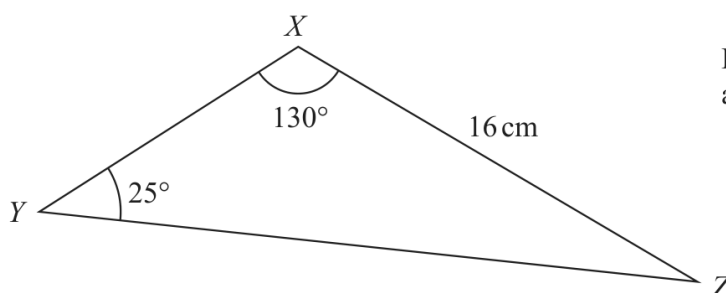


Diagram **NOT**
accurately drawn

The length XZ and the angles YXZ and XYZ are each given correct to 2 significant figures.

Calculate the upper bound for the length YZ .

Give your answer correct to one decimal place.

Show your working clearly.

..... cm

(Total for Question 13 is 3 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q18

CH 1 • TOPIC 9 3 marks

18 $A = w - \frac{x^2}{y}$

$w = 3.45$ correct to 2 decimal places.

$x = 1.9$ correct to 1 decimal place.

$y = 5$ correct to the nearest whole number.

Work out the lower bound of the value of A

Show your working clearly.

(Total for Question 18 is 3 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q22

CH 1 • TOPIC 9 4 marks

22 The diagram shows triangle ABC

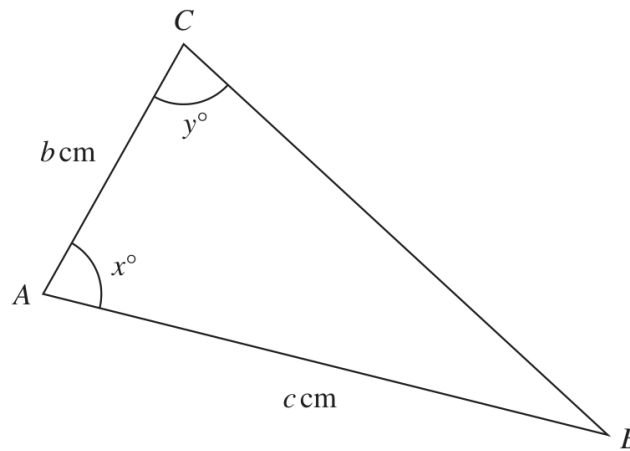


Diagram **NOT**
accurately drawn

$c = 11.5$ correct to one decimal place
 $x = 80$ correct to the nearest whole number
 $y = 75$ correct to the nearest whole number

Calculate the upper bound for the value of b

Show your working clearly.

Give your answer correct to 3 significant figures.

(Total for Question 22 is 4 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q18

CH 1 • TOPIC 9 3 marks

$$18 \quad X = \frac{2a - b}{f}$$

 $a = 7.5$ correct to 1 decimal place. $b = 3.42$ correct to 2 decimal places. $f = 2$ correct to the nearest whole number.Work out the upper bound of the value of X

Show your working clearly.

(Total for Question 18 is 3 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q17

CH 1 • TOPIC 9 3 marks

$$17 \quad P = a(c + y)$$

 $a = 8.3$ correct to 2 significant figures $c = 2$ correct to 1 significant figure $y = 15$ correct to the nearest 5Work out the upper bound for the value of P

Show your working clearly.

(Total for Question 17 is 3 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q19

CH 1 • TOPIC 9 3 marks

19 $G = \frac{c}{2f - 3h}$

$c = 8$ correct to the nearest whole number

$f = 6.62$ correct to 2 decimal places

$h = 1.2$ correct to 1 decimal place

Work out the lower bound for the value of G

Give your answer correct to 3 decimal places.

Show your working clearly.

.....

(Total for Question 19 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q21

CH 1 • TOPIC 9 3 marks

21 $T = \frac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.

Show your working clearly.

(Total for Question 21 is 3 marks)

04 Bounds in Context: Decide

“Could it ...?” and “definitely ...?” are decided by the extreme value. “Suitable degree of accuracy” — round both bounds until they agree.

WHEN YOU SEE → YOU DO

“could the speed have been greater than 80?”

→ work out the **upper** bound, compare

“does he definitely have enough?”

→ worst case: least supply, most needed

“by considering bounds ... suitable degree of accuracy”

→ find both bounds, round both until they match

“greatest number of spheres / panels”

→ whole numbers: round down (or up) the division

Upper bound

Lower bound

Round both

Same? That is the answer

WORKED MODEL • NOVEMBER 2025 P2H

$D = \frac{n}{p - q}$, $n = 10.3$ (1 d.p.), $p = 7.24$, $q = 4.39$ (2 d.p.). Find D to a suitable degree of accuracy.

$$1 \quad \text{UB} = \frac{10.35}{7.235 - 4.395} = 3.644$$

biggest

$$2 \quad \text{LB} = \frac{10.25}{7.245 - 4.385} = 3.584$$

smallest

ANSWER $D = 3.6$ (both bounds round to 3.6 to 2 s.f.)

IDEA 04

Practise: Bounds in Context: Decide

My questions first, then selected practice, then real past papers.

Q34 DR ESLAM Could it?

CH 1 • TOPIC 9 3 marks

A car travels 120 km (nearest km) in 1.5 hours (1 d.p.). Could its average speed have been more than 82 km/h?

SELECTED PRACTICE**Q35** PRACTICE Practice • Hard

CH 1 • TOPIC 9 4 marks

- 1 Dan does an experiment to find the value of π .
He measures the circumference and the diameter of a circle.

He measures the circumference, C , as 170 mm to the nearest millimetre.

He measures the diameter, d , as 54 mm to the nearest millimetre.

Dan uses $\pi = \frac{C}{d}$ to find the value of π .

Calculate the upper bound and the lower bound for Dan's value of π .

(4 marks)

Q36**PRACTICE**

Practice • Hard

CH 1 • TOPIC 9**4 marks**

13 Edith's van can safely carry a maximum load of 920 kilograms.

She wants to use her van to carry

30 sacks of potatoes, each of mass 25 kilograms to the nearest kilogram

and

20 sacks of carrots, each of mass 7.5 kilograms to 1 decimal place.

Can she definitely use her van safely in one journey?

You **must** show your working.

(4 marks)

Q37**PRACTICE**

Practice • Hard

CH 1 • TOPIC 9 4 marks

17 A solid metal cube has sides of length 125 mm, correct to 3 significant figures.

The cube is melted down and the metal used to make solid spheres.

The volume of each sphere is to be 140 cm^3 , correct to the nearest 10 cm^3

Work out the greatest number of spheres that could be made from the metal.

Show your working clearly

(4 marks)

Q38**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 9**3 marks**

- 8 A £1 coin weighs 8.75 g, correct to the nearest 0.01 g. Mitul weighs the contents of a large bag of £1 coins. The coins weigh 2.63 kg, correct to the nearest 10 g.

Mitul says

I am sure that the bag contains exactly £300 because, using bounds,
 $2625 \div 8.755 = 299.8$ to 1 decimal place.

Show that Mitul may not be correct.

[3]

(3 marks)

PAST PAPERS

Q39**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q17

CH 1 • TOPIC 9 4 marks

- 17 A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3}) \text{ m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

..... kg/m^3

(Total for Question 17 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q18

CH 1 • TOPIC 9 4 marks

18 Kaidan and Sonja went on two different car journeys.

For Kaidan's journey

distance = 80 km correct to the nearest 5 km

time = 2.7 hours correct to 1 decimal place

For Sonja's journey

distance = 33 km correct to 2 significant figures

time = 1 hour correct to the nearest 0.1 hour

Kaidan says,

“My average speed could have been greater than Sonja's average speed.”

By considering bounds, show that Kaidan is correct.

Show your working clearly.

(Total for Question 18 is 4 marks)

Q41**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q25

CH 1 • TOPIC 9 4 marks

- 25** A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.
Give your answer in g/cm^3 correct to 2 decimal places.
Show your working clearly.

..... g/cm^3

(Total for Question 25 is 4 marks)

Q42**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q18

CH 1 • TOPIC 9 4 marks

18 Diego builds a fence using fence panels.

The total length of the fence is 50 metres, correct to the nearest 5 metres.

The length of each fence panel is 1.3 metres, correct to the nearest 10 cm.

The cost of each fence panel is £8.65

Diego may only buy complete fence panels.

Diego only pays for the number of panels he needs to build the fence.

Work out the greatest difference in the possible amounts that Diego could pay to build the fence.

Show your working clearly.

£.....

(Total for Question 18 is 4 marks)

Q43**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q19

CH 1 • TOPIC 9 3 marks

19 Aviv goes on a cycle journey.

For the cycle journey

average speed = 19 km/h correct to the nearest whole number

time = 1.5 hours correct to one decimal place

Work out the upper bound for the distance Aviv travels.

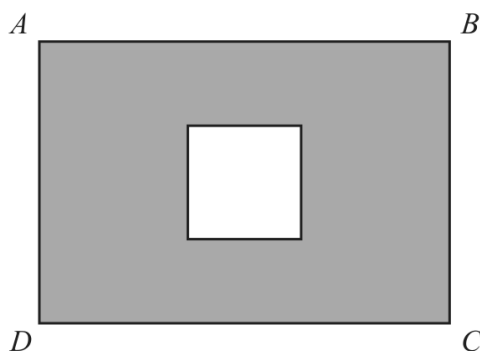
Give your answer correct to 3 significant figures.

..... km

(Total for Question 19 is 3 marks)

Q44 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 2H • Q22

CH 1 • TOPIC 9 4 marks

22 The diagram shows a square inside rectangle $ABCD$ Diagram **NOT**
accurately drawnThe total area of the region shown shaded in the diagram is $X \text{ cm}^2$ $AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm $BC = 9.2 \text{ cm}$ correct to 2 significant figuresside of square = 4.1 cm correct to 2 significant figuresBy considering bounds, work out the value of X to a suitable degree of accuracy.

Show your working clearly.

 $X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

Q45**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q24

CH 1 • TOPIC 9 5 marks

24 $D = \frac{n}{p - q}$

 $n = 10.3$ correct to 1 decimal place $p = 7.24$ correct to 2 decimal places $q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

.....
(Total for Question 24 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE BIGGEST

UP on top, LOW below

UP – LOW

ZONE SMALLEST

LOW on top, UP below

LOW – UP

- | | | |
|-----------|------------------------|--|
| 01 | Error intervals | Half the unit each way: $LB \leq x < UB$. |
| 02 | Estimate | Everything to 1 s.f.; nice squares under roots. |
| 03 | Formula bounds | Choose each letter to push the answer the way you want. |
| 04 | Decide | Extreme case for “could / definitely”; round both bounds to agree. |

BOTTOM LINE Unit → half → bounds → for each letter ask: does this make my answer bigger or smaller?

Answers

Rounding, Estimation and Bounds • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • ROUNDING AND ERROR INTERVALS

- | | | |
|------------|---|---|
| Q1 | Dr Eslam • Half the unit | $6.35 \leq w < 6.45$ |
| Q2 | Practice • Easy | $4.755 \leq n < 4.765$ |
| Q3 | Practice • Easy | 33.75 cm, 33.85 cm |
| Q4 | Practice • Easy | (a) $8.35 \leq l < 8.45$ (b) $41.75 \leq P < 42.25$ |
| Q5 | Practice • Medium | (a) $5.5 \leq l < 6.5$ (b) e.g. $6.4 + 4.6 = 11$ |
| Q6 | Edexcel 4MA1 • January 2020 • Paper 1HR • Q11 | (a) 4.35 kg (b) 4.25 kg (c) $17.5 - 9.35 = 8.15$ |
| Q7 | Edexcel 4MA1 • January 2022 • Paper 2H • Q4 | (a) 17.75 mm (b) 18.25 mm |
| Q8 | Edexcel 4MA1 • November 2024 • Paper 2H • Q2 | (a) 1.45 m (b) 1.35 m |
| Q9 | Edexcel 4MA1 • June 2025 • Paper 1HR • Q2 | (i) 142.75 m (ii) 142.85 m |
| Q10 | Edexcel 4MA1 • November 2025 • Paper 1H • Q7 | (a) 474.5 g (b) 125 cm |

IDEA 02 • ESTIMATION

Q11 Dr Eslam • 1 s.f.

$$\frac{50 \times 4}{0.2} = 1000$$

Q12 Practice • Easy

$$\frac{300 \times 4}{60} = 20: \text{reasonable}$$

Q13 Practice • Medium

$$\frac{70 \times 40}{0.02} = 140\,000: \text{sensible}$$

Q14 Practice • Hard

$$\frac{90\,000}{2000} \approx 45$$

Q15 Practice • Very Hard

(a) $\frac{1000}{4 \times 10^{-22}} = 2.5 \times 10^{24}$ (b) underestimate:
the mass was rounded up

Q16 Edexcel 4MA1 • November 2025
• Paper 1H • Q2

$$\frac{2^2 \times 60}{3} = 80$$

IDEA 03 • BOUNDS IN A FORMULA

- Q17** Dr Eslam • Biggest $\frac{12.5}{3.35} = 3.73$
- Q18** Practice • Medium (a) 4.25 (b) $4.35 + \frac{1}{0.35} = 7.21$
- Q19** Practice • Medium $\frac{99.75}{66.5} = 1.5$
- Q20** Practice • Hard $\sqrt{8.35^2 - 6.05^2} = 5.75 \text{ cm}$
- Q21** Practice • Very Hard $\frac{347.5}{2.55 \times 6.35} = 21.46$
- Q22** Edexcel 4MA1 • January 2020 • Paper 1H • Q16 $\frac{2(58.45) - 19.5}{3.55} = 27.44$
- Q23** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q19 $8.35 \times 7.25 - 6.15 \times 5.25 = 28.25 \text{ cm}^2$
- Q24** Edexcel 4MA1 • January 2021 • Paper 1H • Q15 $\frac{9.65 - 3.75}{1.835} = 3.22$
- Q25** Edexcel 4MA1 • January 2021 • Paper 1HR • Q19 $\frac{13.5}{7.85 - 3.35} = 3$
- Q26** Edexcel 4MA1 • May 2021 • Paper 1H • Q20 $\frac{9.75 - 5.85}{2.5} = 1.56$
- Q27** Edexcel 4MA1 • November 2021 • Paper 1H • Q13 $\frac{16.5 \sin 125^\circ}{\sin 24.5^\circ} = 32.6 \text{ cm}$
- Q28** Edexcel 4MA1 • January 2022 • Paper 1H • Q18 $3.445 - \frac{1.95^2}{4.5} = 2.6$
- Q29** Edexcel 4MA1 • January 2022 • Paper 1HR • Q22 $\frac{11.55 \sin 26^\circ}{\sin 74.5^\circ} = 5.25 \text{ cm}$
- Q30** Edexcel 4MA1 • June 2022 • Paper 2H • Q18 $\frac{2(7.55) - 3.415}{1.5} = 7.79$
- Q31** Edexcel 4MA1 • November 2023 • Paper 2H • Q17 $8.35(2.5 + 17.5) = 167$

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- Q32** Edexcel 4MA1 • June 2024 • Paper 2H • Q19 $\frac{7.5}{2(6.625) - 3(1.15)} = 0.765$

IDEA 04 • BOUNDS IN CONTEXT: DECIDE

- Q34** Dr Eslam • Could it? $UB = \frac{120.5}{1.45} = 83.1 > 82$: yes
- Q35** Practice • Hard $UB \frac{170.5}{53.5} = 3.19$, $LB \frac{169.5}{54.5} = 3.11$
- Q36** Practice • Hard $\max 30(25.5) + 20(7.55) = 916 \leq 920$: Yes
- Q37** Practice • Hard $\frac{125.5^3 \div 1000}{135} = 14.6$: 14 spheres
- Q38** Practice • Very Hard bounds $\frac{2625}{8.755} = 299.8$ to $\frac{2635}{8.745} = 301.3$:
could be £301
- Q39** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q17 $\frac{5.025}{1.835 \times 10^{-3}} = 2738.4 \text{ kg/m}^3$
- Q40** Edexcel 4MA1 • June 2022 • Paper 2HR • Q18 Kaidan $UB \frac{82.5}{2.65} = 31.1 > 31.0$ Sonja $LB \frac{32.5}{1.05} =$
- Q41** Edexcel 4MA1 • June 2023 • Paper 2H • Q25 $\frac{265}{\frac{4}{3}(2.75)^3} = 9.56 \text{ g/cm}^3$
- Q42** Edexcel 4MA1 • May 2023 • Paper 1HR • Q18 $\max 42 \text{ panels, min } 36: 6 \times 8.65 = \text{£}51.90$
- Q43** Edexcel 4MA1 • June 2024 • Paper 2HR • Q19 $19.5 \times 1.55 = 30.2 \text{ km}$
- Q44** Edexcel 4MA1 • June 2025 • Paper 2H • Q22 $UB 92.3, LB 85.7: X = 90 (1 \text{ s.f.})$
- Q45** Edexcel 4MA1 • November 2025 • Paper 2H • Q24 $UB 3.64, LB 3.58: D = 3.6$